

Linear Regression from Scratch

Predicting CO₂ Emissions with Pure
Mathematics

Math 204 — Linear Algebra for Computer Science

German International University in Egypt

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Why CO₂ Emissions Matter —

Transport accounts for a major share of global emissions; predicting vehicle CO₂ supports sustainability decisions.

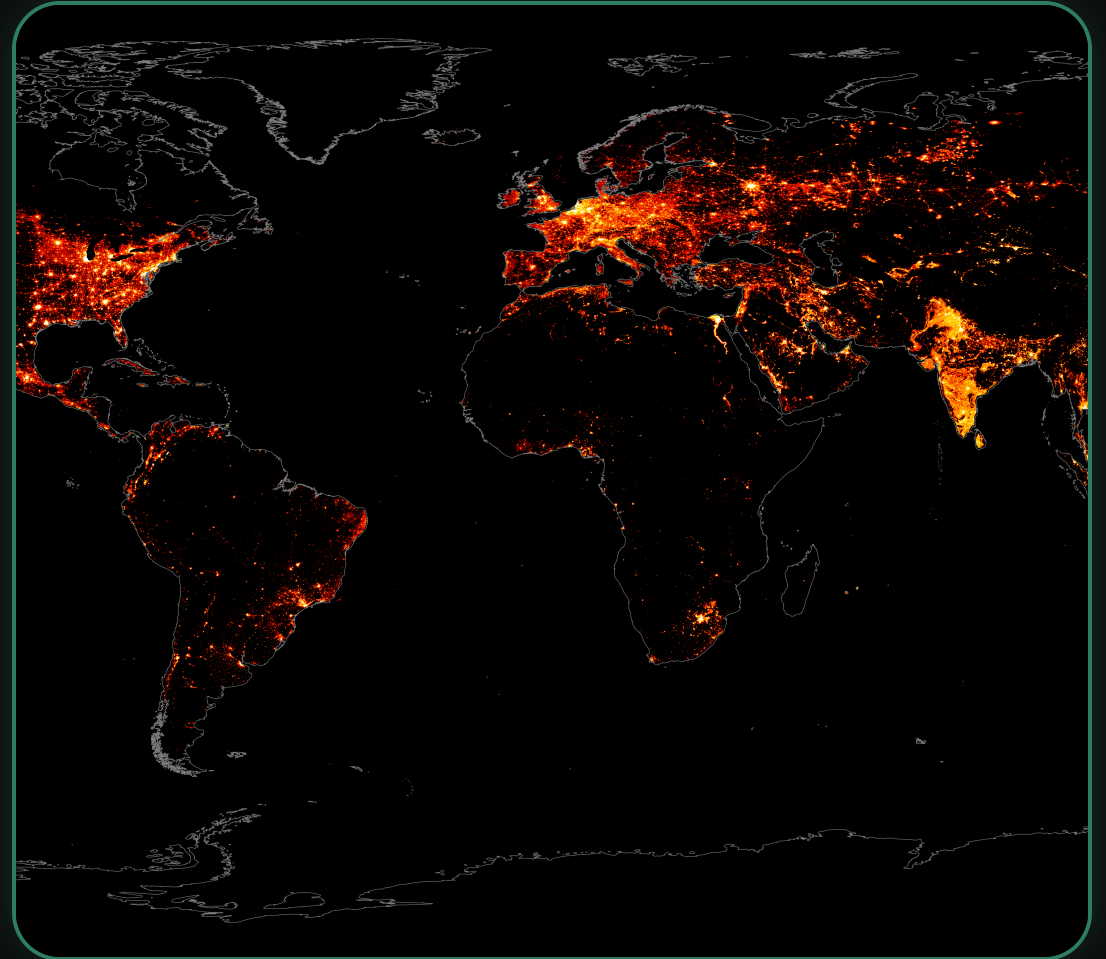


Emissions depend on engine characteristics



Prediction enables policy/standards, consumer awareness, and fleet planning

Vehicle specs → CO₂ prediction model → Decision impact
(policy / product / consumer)



Real-World Data: Canadian Vehicles



Dataset

~7,385 vehicle records

x

Features

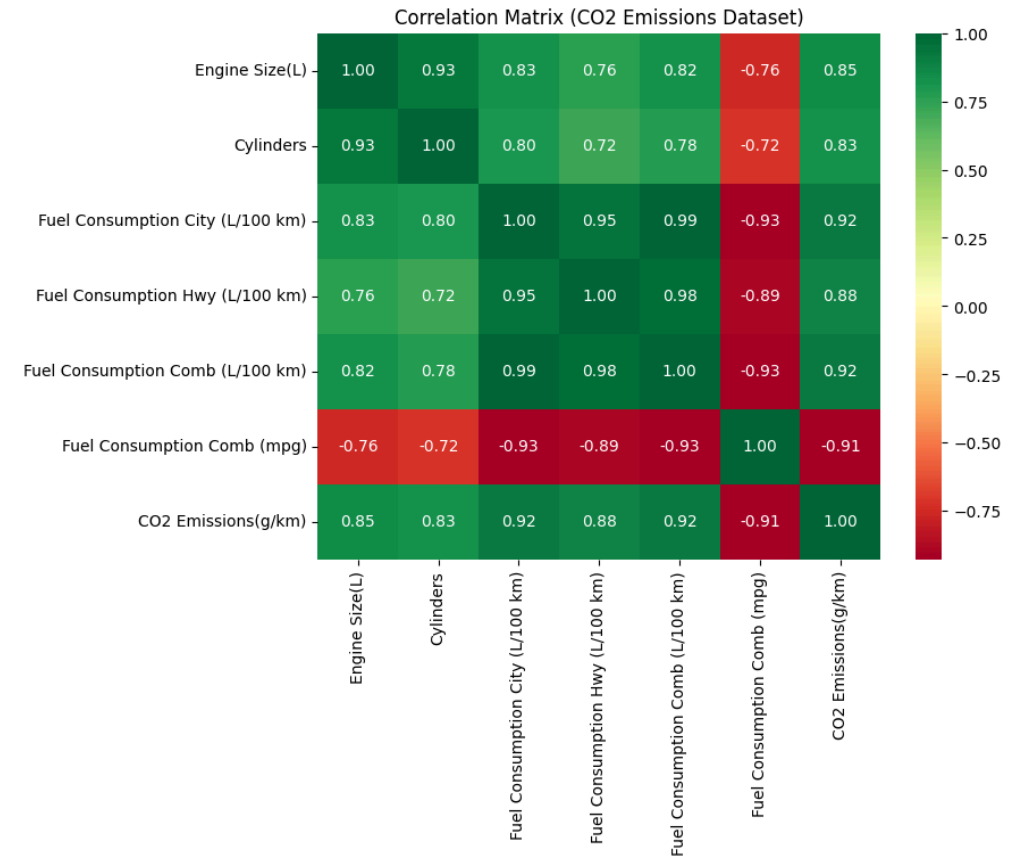
Engine Size (L), Cylinders

y

Target

CO₂ emissions (g/km)

Two intuitive features can explain a meaningful portion of emission variance.



Linear Algebra Foundations —



Vectors & Matrices

Representing data as **X** (features), **y** (targets), and **w** (weights).



Minimizing Error

Finding the optimal weights to reduce the difference between predictions and actual values.



Least Squares Concept

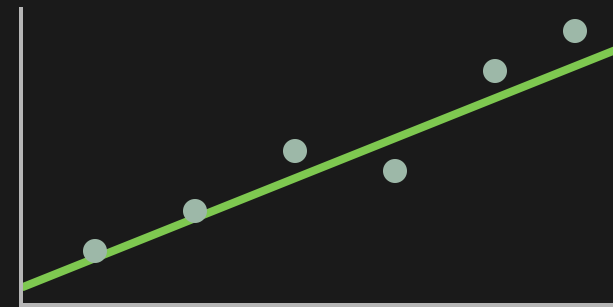
Minimizing the sum of squared residuals to find the best-fitting line.

Matrix Representation

$$y = X w + b$$

Target = Features × Weights + Bias

Data → Line Fit



Normal Equation —



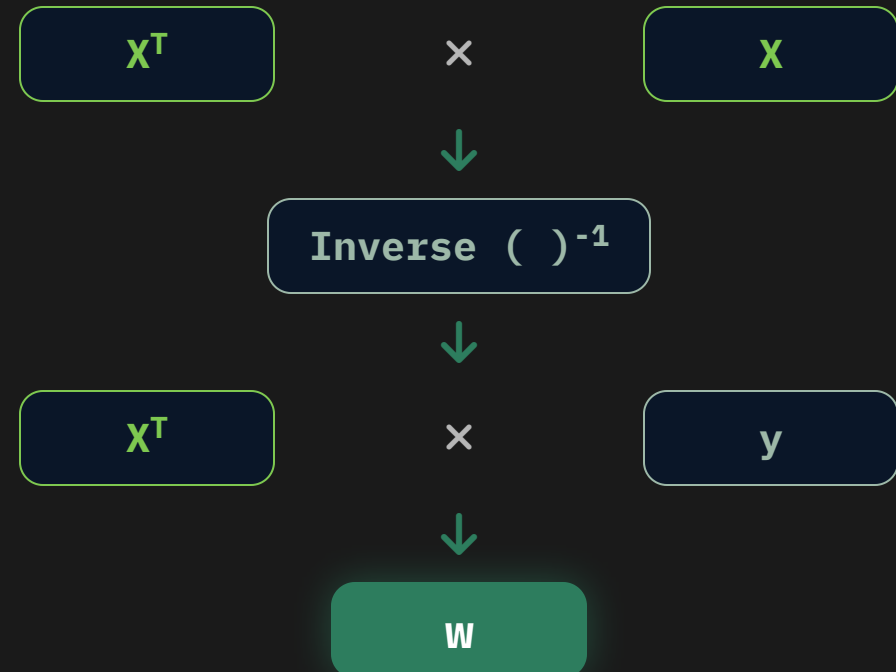
Direct closed-form solution

Compute once, get optimal weights.

Formula

$$W = (X^T X)^{-1} X^T y$$

Matrix Multiplication Flow



Gradient Descent —

Formula

$$w = w - \alpha \cdot X^T(Xw - y)$$



Iterative Optimization

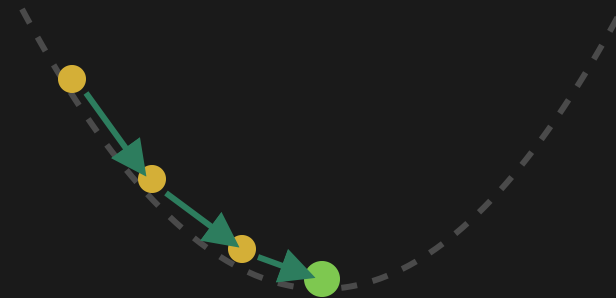
Gradually updates weights step-by-step to find the minimum error.



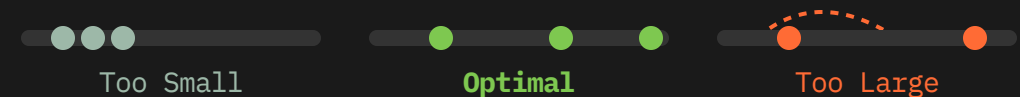
Key Parameters

Requires tuning the **learning rate (α)** and setting the number of iterations for convergence.

Step-by-Step Descent



Learning Rate (α) Impact



Ridge Regression —



L2-Regularized Solution

Adds a penalty term to the loss function to constrain weight magnitudes and improve model stability.

Formula

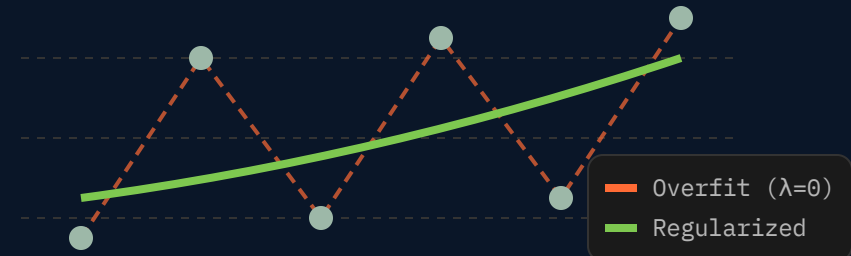
$$W = (X^T X + \lambda I)^{-1} X^T y$$



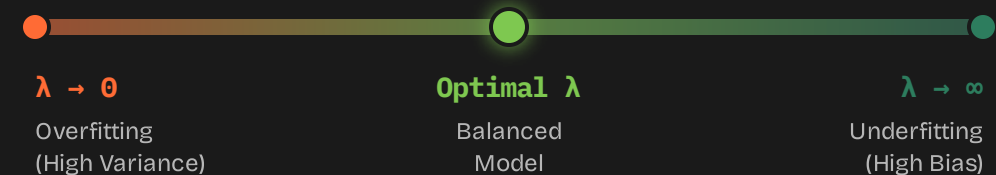
Prevents Overfitting

The regularization parameter λ controls the trade-off between fitting training data and keeping weights small.

Regularization Effect



λ Tuning Impact



Engineering the Solution

Tools



NumPy

(core math)



pandas

(data handling)



matplotlib

(plots)

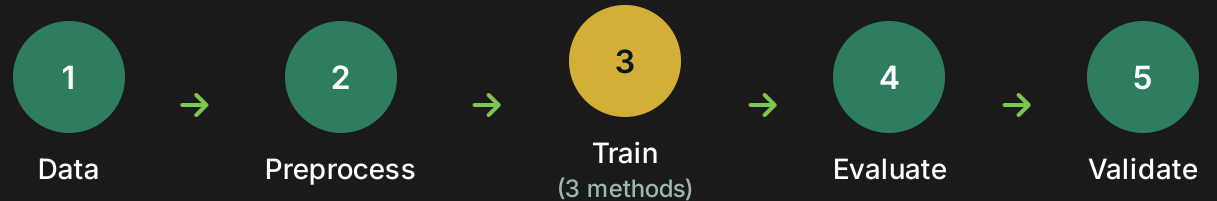


sklearn

(validation only)

Key decisions

- Prefer `np.linalg.solve()` over explicit inverse
- Normalize features for gradient descent
- Exclude bias term from Ridge regularization



All Methods Agree

R² SCORE

≈ 0.72–
0.73

MSE

≈ 900

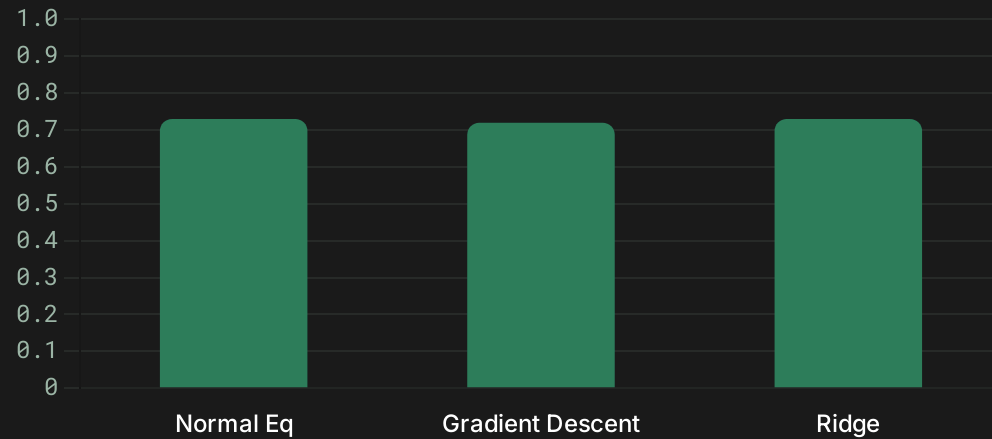
PREDICTIONS WITHIN

~30 g/km

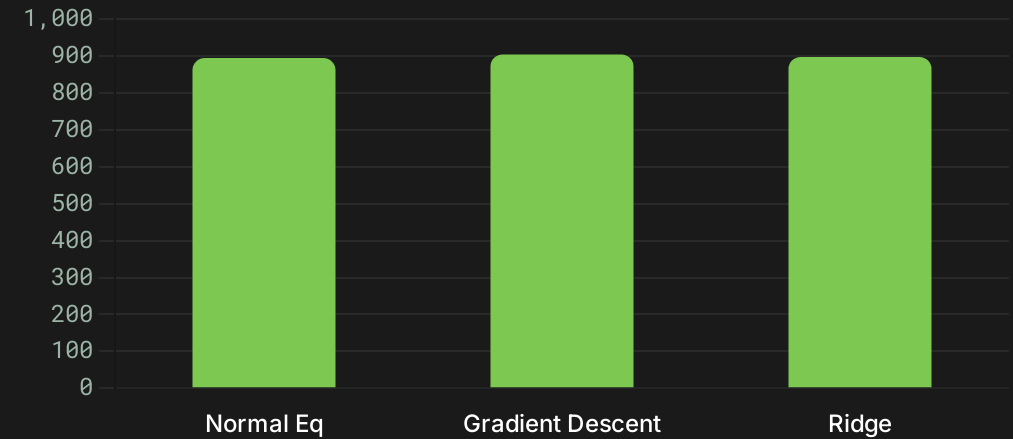
WEIGHT DIFF VS SKLEARN

≈ 10⁻¹²

R² by Method



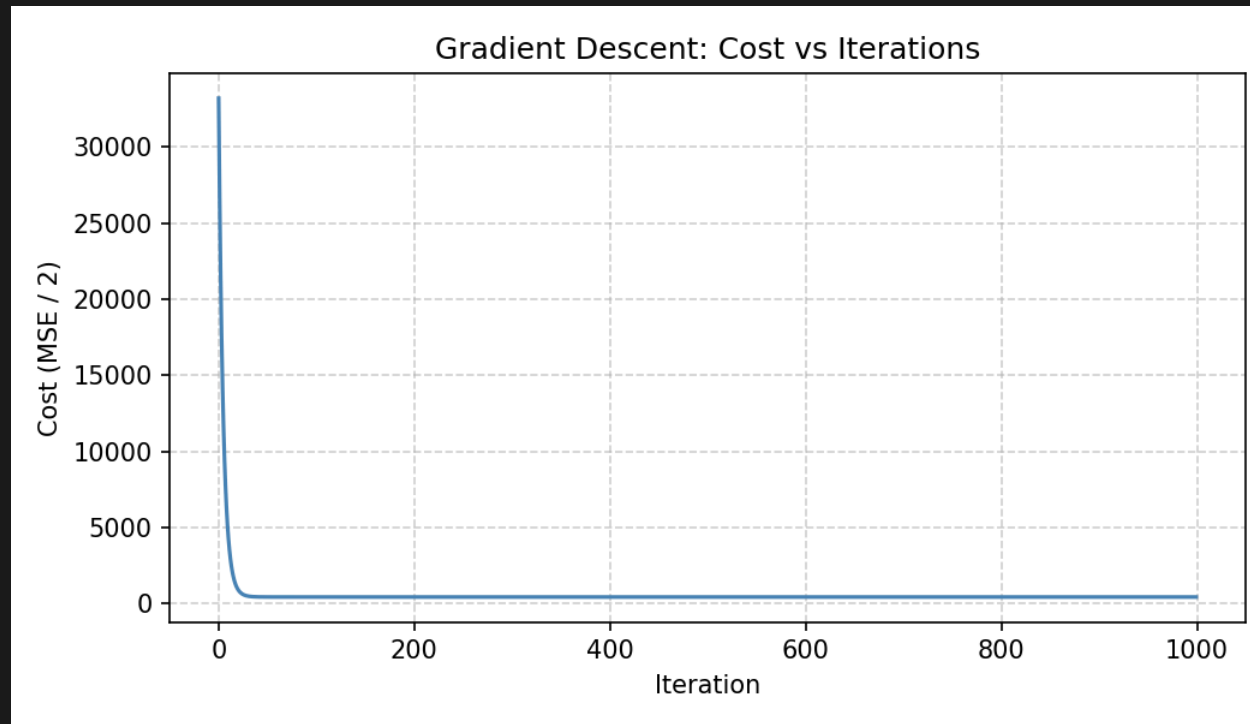
MSE by Method



Watching the Model Learn —

Rapid error drop, then stable convergence.

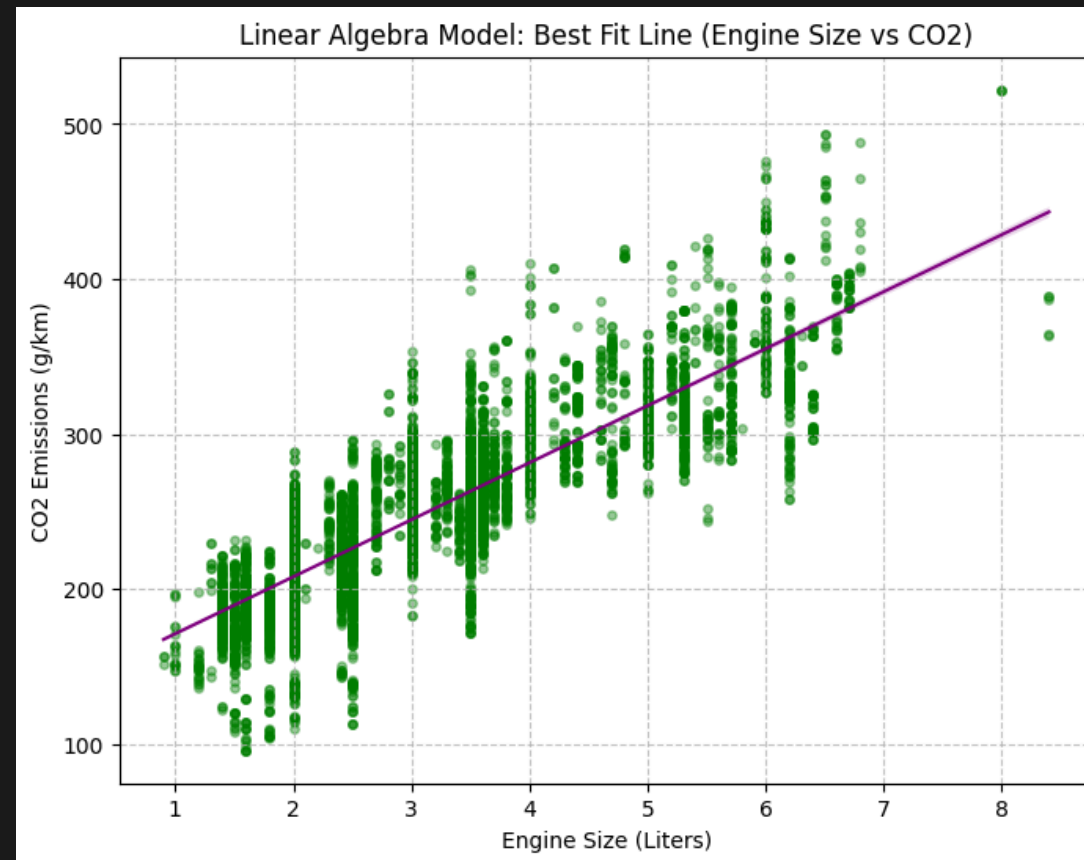
● Rapid learning phase



● Convergence plateau

From Math to Reality

- Insight: A simple linear model produces practical accuracy ($\approx \pm 30$ g/km).



What We Learned —



Mathematical Truth

Linear algebra fundamentals
are sufficient to build
regression end-to-end



Practical Insight

From-scratch weights match
sklearn ($\approx 10^{-12}$ difference)



Real-World Impact

Emission predictions support
sustainability decisions

 **Green AI impact**

Thank You

Questions?



Ahmed Roshdy



Khalid Osam



Omar Shaker



Yousef Hany



Mark Tamer



Hassan Eslam



Carlos Emad



Mohamed Elsayed



Ahmed Fouad